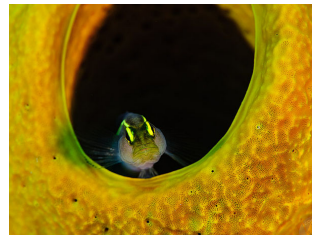


Take a deep breath. Relax. Its just 50 minutes of your life. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points.

1. Competition occurs between two different species in a community when
 - A. one species is a parasite of the other.
 - B. both species need the same resources that is in limited supply.
 - C. both species have partitioned resources.
 - D. both species are searching for mates.
 - E. one species is a predator of the other.
2. Some fish, like the goby in the picture below, live inside sponges. Sponges are a type of living organism. The goby benefits from the shelter provided by the sponge. The sponge receives no apparent benefit. As described here, this is an example of

- A. parasitism.
- B. predation.
- C. symbiosis.
- D. commensalism.
- E. mutualism.

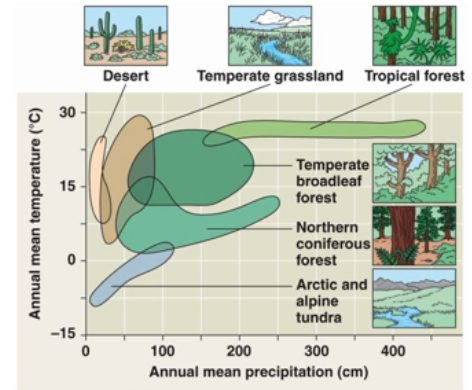


3. If emigration is greater than immigration and if the death rate is greater than the birth rate then
 - A. the population size will decrease.
 - B. the population size will get larger if the birth rate is larger than immigration.
 - C. the population has plenty of resources.
 - D. the population size will remain about the same.
 - E. the population size will increase by exponential growth.
4. Consider the food chain grass → grasshopper → mouse → snake → hawk. How much of the chemical energy fixed by photosynthesis of the grass (100%) is available to the hawk?
 - A. 1%
 - B. 0.1%
 - C. 10%
 - D. 0.01%
 - E. 60%

5. After transient orca began to prey on sea otters, what happened to the kelp forest?
- A. The kelp abundance remained about the same. Only the predator-prey interactions between Orca and otters were important, so nothing happened to the kelp.
 - B. The kelp abundance decreased. Sea otters and kelp have a symbiotic relationship. Without the sea otters, the symbiotic relationship was broken and so the kelp abundance decreased.
 - C. The kelp abundance decreased. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.
 - D. The kelp abundance remained about the same. The kelp are not part of this community, so the kelp could not be affected.
 - E. The kelp abundance increased. Otters dislodge kelp with their playful antics. Because there were fewer otters, less kelp was dislodged so the kelp abundance increased.
6. In ecosystems, why is the “cycling” used to describe the transfer of chemical elements and nutrients, but the “flow” is used for the transfer of energy?
- A. Energy flows from one ecosystem to other ecosystems, but chemical elements constantly cycle within an ecosystem.
 - B. Both chemical elements and energy flow within an ecosystem but can cycle to other ecosystems.
 - C. Both chemical elements and energy are recycled but can flow to other ecosystems.
 - D. Chemical elements are recycled within ecosystems, but energy flows through and out of ecosystems.
 - E. Chemical elements are cycled into ecosystems from other ecosystems, but energy constantly flows within the ecosystem.
7. Assuming a population has unlimited resources, exponential growth occurs because
- A. the immigration rate equals the death rate and the emigration rate equals the birth rate.
 - B. the birth rate exceeds the death rate.
 - C. the emigration rate exceeds the death rate.
 - D. the emigration exceeds the birth rate.
 - E. the death rate exceeds the birth rate.
8. The energy that is *not* transferred to higher trophic levels is lost
- A. to secondary consumers.
 - B. to primary consumers.
 - C. to decomposers.
 - D. as metabolic heat.
 - E. in fecal matter.
-

9. The southeastern region of Missouri lies within the temperate broadleaf forest ecosystem. According to the climograph below, the temperate broadleaf forest

- A. never has more than about 220 cm (87 inches) of rainfall per year.
- B. always has more precipitation than the temperate grassland.
- C. has more rain and similar temperatures to the northern coniferous forest.
- D. has an average of 60–220 cm of precipitation per year.
- E. is never warmer than the temperate grassland.



10. When the number of individuals in a population remains more or less the same over time in the specific place where these individuals live, then this population has reached the _____ of that place.

- A. maximum growth capacity
- B. carrying capacity
- C. population density
- D. logistic growth capacity
- E. exponential growth capacity

11. Carrying capacity

- A. is the maximum population size that a particular environment can support.
- B. is determined by birth and death rates, and emigration and immigration rates.
- C. describes the stress a population undergoes due to limited resources.
- D. does not for most species most of the time.
- E. is seldom reached by most population because the environment has plenty of resources.

12. On average, the amount of secondary consumer biomass is about _____ less than the amount of primary producer biomass.

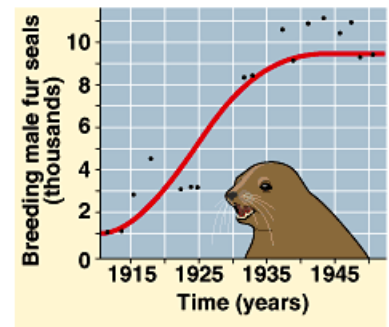
- A. 20×
- B. 50×
- C. 10×
- D. 100×
- E. 80×

13. Which of the following is *not* one of the four factors that play a role in population growth rate?

- A. Emigration.
- B. Immigration.
- C. Birth rate.
- D. Death rate.
- E. Demography.

14. The type of growth curve shown in the figure below is a

- A. maximum growth curve.
- B. exponential growth curve.
- C. carrying capacity growth curve.
- D. logistic growth curve.
- E. population density growth curve.



15. Considering your knowledge of the dead zone and the causes of microscopic algae blooms, the Gulf of Mexico ecosystem is emphmost likely to be limited by which of the following nutrients?

- A. oxygen
- B. hydrogen
- C. carbon
- D. CO₂
- E. nitrogen

16. Which of the following statements about symbiotic relationships is true?

- A. A relationship that appears to be commensalism may in fact be mutualism or parasitism.
- B. Both organisms are harmed in a parasitic relationship.
- C. Neither organism benefits in a symbiotic relationship.
- D. Both symbiotic organisms usually benefit from the relationship.
- E. The most efficient type of parasite is one that kills its host.

17. Probably the most important factor(s) affecting the distribution of ecosystems is (are)

- A. day length and rainfall.
- B. longitude and latitude.
- C. wind and ocean water current patterns.
- D. climate.
- E. wind direction and rain shadows.

18. The land on the south side of the Himalayan mountain range (below) receives relatively high levels of rainfall and has fertile farmlands. The land north of the mountain range is hot and dry and can't be farmed. Which of the following is true.

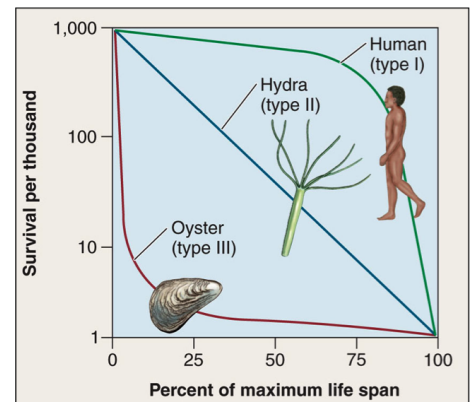
- A. The cool, moist Indian Ocean air heats up as it rises, releasing its precipitation as it passes the tops of the mountains, and this warm, now dry air cools as it descends on the leeward side of the range.
- B. The south side of the Himalayan mountain range is in a rain shadow.
- C. Warm, moist air from the Indian Ocean blows north, rises and cools, releasing precipitation as it moves up the south side of the range, and this cool, now dry air mass heats up as it descends on the north side of the range.
- D. The warm, moist air is too dense to rise over the height of Mount Everest and the other mountains in this range so the moisture remains on the south side of the range.
- E. The desert in southern China warms the north side of the mountain range, causing it to dry out.



19. The dead zone in the Gulf of Mexico and other oceans around the globe occurs when
- A. an excess of nutrients causes an algal bloom. The algae consume all of the oxygen in the water for photosynthesis, so other organisms can no longer live there.
 - B. an excess of nutrients causes an algal bloom, followed by death of the algae. Decomposition of the algae uses up the oxygen in the water, so other organisms can no longer live there.
 - C. commercial fishers overharvest shrimp, crab, fishes and other sea animals.
 - D. global climate change causes the water to become too warm to sustain life.
 - E. eutrophication causes an absence of nutrients in the zone.
20. The difference between communities and ecosystems is that
- A. communities include individuals of the same species; ecosystems include individuals of different species.
 - B. communities include individuals of the different species; ecosystems are individuals of the same species.
 - C. communities do *not* include interactions between individuals and the physical environment; ecosystems *do* include such interactions.
 - D. communities are groups of individuals of the different species; ecosystems are groups of communities.

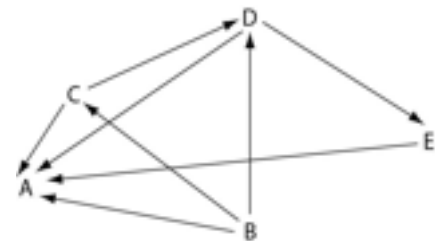
21. If humans stopped fishing along the coast of Alaska, what will probably happen to the kelp forest and why?
- The kelp abundance will increase because of the symbiotic relationship between kelp and fishes. Fishes are necessary for kelp to reproduce.
 - The kelp abundance will remain about the same. The kelp are not part of this community, so the kelp would not be affected.
 - The kelp abundance will decrease. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.
 - The kelp abundance will decrease because more fish will be present to eat the kelp.
 - The kelp abundance will increase because the sea urchin abundance will probably decrease.
22. According to the figure below, which is most correct about the offspring produced by oysters?

- About the same number of offspring will die every year.
- Most offspring will die at a very young age.
- Most offspring will live most of their natural lifespan.
- Most offspring will die late in life.
- Most offspring will die about half way through their natural life.



23. Several species of darters (a group of small fishes) live together on the gravel bottoms of small streams. The darters can live in the same habitat type because they
- can swim at different heights above the gravel bottom.
 - feed at different times of the day.
 - created new niches within the stream.
 - divided up the resources to reduce competition.
 - outcompeted other species of fishes, such as catfishes.
24. Based on this food web, which letter represents an organism that is a primary producer? The arrows point in the direction of energy flow (to higher trophic levels).

- C
- D
- E
- A
- B



25. The “role” or “job” of a species in a community, represented by all of the resources needed by the species, is called
- A. a predator or prey.
 - B. the niche.
 - C. a predator.
 - D. the resource.
 - E. a producer or consumer.
26. Which of the following are important biotic (living) factors that can affect the organization of communities?
- A. predation, competition
 - B. nutrient availability, soil pH
 - C. precipitation, wind
 - D. light intensity, seasonality
 - E. temperature, water
27. The difference between emigration and immigration is
- A. the direction in which individuals are moving into or out of the population.
 - B. the overall movement of individuals among populations.
 - C. emigration is when individuals move out of a population; immigration is when individuals move into a population.
 - D. emigration is when individuals move out of a population; immigration is when individuals move out of a community.
 - E. emigration is when individuals move into a population; immigration is when individuals move out of a population.
28. In 2011, the Mississippi River experienced a record flood event. Which of the following statements is the best prediction of how this record flood event will affected the Gulf of Mexico dead zone that summer?
- A. The dead zone will be smaller because the volume of water will dilute the nutrients, reducing eutrophication.
 - B. The dead zone will be larger because the volume of water will cause more fishes to avoid the freshwater input.
 - C. The dead zone will be larger because the volume of water will carry even more nutrients into the Gulf of Mexico.
 - D. The dead zone will be smaller because the volume of water will bring more oxygen to the Gulf of Mexico, which will offset the hypoxia.
 - E. The dead zone will be similar to previous years because most of the flood water will soak into the ground and not enter the Gulf of Mexico.
-

29. Higher elevations on a mountain experience less precipitation and cooler temperatures than low elevations. As you descended from the top of the mountain, you would notice a change in the types of plants and animals present in the community. This is analogous to the changes in the ecosystems if you walked
- A. from the north pole to the equator.
 - B. east to west over different altitudes.
 - C. from the north pole to the south pole.
 - D. from the equator to the south pole.
 - E. west to east at one latitude.
30. Plants produce water vapor during photosynthesis, which goes from the plant to the atmosphere. The water vapor in the atmosphere eventually falls back to the earth as rain. This process would be part of
- A. plant metabolism.
 - B. the water cycle.
 - C. the entire ecosystem.
 - D. energy flow.
 - E. the energy cycle.
-